More Atoms and Ions Practice

1. What does the atomic number tells you about an atom?

2. What is the atomic mass of an atom?

3. If an atom has 13 protons and 14 neutrons in its nucleus, what is its atomic mass?

4. An atom has an atomic mass of 91 and an atomic number of 40. How many neutrons are there in its nucleus?

5. An atom has an atomic mass of 55 and has 30 neutrons in its nucleus. How many protons does the atom have?

6. Draw a labelled diagram of an atom of fluorine, with the electrons in shells.

7. Complete the following table using your periodic table.

Element Symbol	Element Name	Atomic Number	Number of Protons	Atomic Weight	Number of Neutrons	Number of Electrons
Pb						
	Potassium					
		30				
			22			
				197		
						56
	Bromine					

8. What part of an atom can be gained or lost by an atom to become an ion?

9. If an atom loses 3 electrons, what will be the charge on the ion?

10. If an atom gains 2 electrons, what will be the charge on the ion?

11. What ending do you need to use when naming a negative ion?

12. Use your periodic table to predict what charge ions of the following elements will have. Name each ion.

Element	Element Symbol	Symbol of Ion	Name of Ion
Lithium			
Oxygen			
Bromine			
Aluminium			
Calcium			
Strontium			
Potassium			
Fluorine			
Magnesium			
Chlorine			

13. Name the following ions and state how many electrons have been lost or gained.

a. Na^{+}
b. Zn^{2+}
c. I^{-}
d. N^{3-}
e. Cu^{2+}
f. S^{2-}

14. What is a polyatomic ion?

15. Write the names of the following polyatomic ions.

a. OH^{-}
b. PO_4^{-3}
c. SO_4^{-2}
d. NO_3^{-}
e. NH_4^{+}
f. CO_3^{-2}

16. Write the symbol for each ion below.

a. Hydrogen
b. Copper (II)
c. Sulphide
d. Iodide
e. Lithium
f. Phosphate
g. Hydroxide
h. Potassium
i. Iron (III)
j. Magnesium
k. Chloride
l. Nitrate
m. Carbonate
n. Zinc
o. Oxide
p. Sulphate
q. Fluoride
r. Calcium
s. Bromide
t. Chromium (III)
u. Lead (IV)